

Crime, Safety, Environment

And The Urban Feasible Set For Work

Special Topic 4: Urban and Labor – Week 4

Central question

- How do safety and environmental conditions change which jobs workers can take and how much work is worth?
- This is a labor lecture because risk changes search, commuting, hours, productivity, retention, sorting, and welfare.
- The relationship is two-sided: urban risk shapes work, and labor-market opportunity shapes crime and risky activity.
- Wages are not enough when risk and exposure differ across workers and places.

Where Week 4 fits

- Week 1: workers choose job-residence bundles, not placeless jobs.
- Week 2: housing and moving costs shape real access.
- Week 3: segregation and neighborhoods create unequal job access.
- Week 4: crime, safety, pollution, heat, and noise alter the feasible set for work.

Risk-adjusted value of work

$$V_{ijrlt} = w_{jlt} - R_{rt} - \tau_{ijrlt} - \rho_{ijrlt}^S - \rho_{ijrlt}^E + A_{rt} + \varepsilon_{ijrlt}$$

Term	Interpretation
w	wage at the job or workplace
R	housing cost tied to residence
τ	generalized commute and schedule cost
ρ^S	safety, harassment, violence, or public-space risk
ρ^E	pollution, heat, noise, or other environmental exposure
A	amenity value

Two-sided equilibrium framework

$$\Pr(C_{ilt} = 1) = F(B_{ilt} - p_{lt}F_i - O_{ilt} + \eta_{ilt})$$

Direction	Labor interpretation
Risk $\rightarrow$ labor	safety and exposure change search, commutes, productivity, hours, retention
Labor $\rightarrow$ crime	job loss and weak earnings prospects lower the value of legal work
Risk $\rightarrow$ sorting	exposed workers and firms change residence, workplace, and sector choices
Risk $\rightarrow$ wages/rents	compensation and capitalization may be incomplete or unequal
Joint equilibrium	observed effects already include worker, firm, and housing responses

Risk and the feasible set

Margin	Labor object	Example outcome
Public safety	search radius, commute timing	employment, accepted wage, commute distance
Workplace harassment	retention, sector choice	turnover, segregation, inequality
Pollution, heat, noise Environmental amenities	productivity, hours, attendance job and residence sorting	output, fatigue, absenteeism compensating differentials, welfare
Hidden harms	welfare versus wages	incomplete compensation

- Local labor-market opportunity changes the opportunity cost of crime and risky activity.
- Job displacement can move earnings, credit stress, and crime risk at the same time.
- Public-space risk changes commute routes, late shifts, job search radius, and schedule feasibility.
- Workplace harassment and violence affect quits, retention, occupational sorting, and inequality.
- The counterfactual must state whether safety, opportunity, or reporting changes.

Environment and labor margins

Exposure	Mechanism	Labor margin
Pollution	health, fatigue, cognition	labor supply, absenteeism, output
Heat	physical stress and time allocation	hours, pace, task choice
Noise	attention and cognitive load	productivity, errors, fatigue
Amenities	sorting and compensation	wages, rents, real welfare

- Exposure is unequal across residence, commute, workplace, task, and bargaining power.
- Adaptation can attenuate estimates while still leaving welfare losses.

Why wages can mislead

- Risk may be hidden, poorly observed, or cumulative.
- Workers may lack mobility or bargaining power to demand compensation.
- Safer places can capitalize into rents rather than worker welfare.
- Firms can change staffing, tasks, or technology instead of wages.
- Equal wages can mask unequal commute risk, harassment, pollution, heat, and stress.

Empirical designs and data

Topic	Typical data	Design family
Crime and opportunity	arrests, layoffs, matched employer data	layoff and demand shocks
Pollution and supply	monitors, satellites, worker panels	plant closures, inversions, panels
Workplace safety	surveys, HR records, complaints	event studies, within-firm designs
Heat, noise, output	weather, noise, task output	high-frequency panel designs
Commute safety	route, transit, crime data	policy and route shocks

- Labor opportunity and crime: local labor markets, disadvantaged youth, job loss, and credit stress.
- Safety as a work constraint: crime timing, street harassment, workplace harassment, sexual violence.
- Environment and productivity: pollution, heat, noise, labor supply, and output.
- Welfare interpretation: compensation is often incomplete, and sorting can hide harms.

- Reproduce: synthetic displacement event study for earnings, opportunity, credit stress, and crime risk.
- Diagnose: classify mechanisms as risk-to-labor, labor-to-crime, sorting, welfare, or equilibrium.
- Transfer: apply the framework to heat, pollution, noise, and worker productivity.
- Extension: add a commute-safety shock that changes feasible shifts while holding wages fixed.

- Crime, safety, and environment are labor-market constraints when they alter feasible work.
- Labor-market opportunity also shapes crime and risky activity.
- Environment matters through labor supply, productivity, health, cognition, and adaptation.
- The best papers name the margin, identify the counterfactual, and avoid treating wages as welfare.